

SE 216 – SOFTWARE PROJECT MANAGEMENT
PROJECT RISKS DOCUMENT

PROJECT NAME: BioPulse : Autonomous Green Ops Platform

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Ata Çelik, Onur Özbek, Enes Görkem Hasgör

LIKELIHOOD RANK	RISK DESCRIPTION
1	Communication Speed and Sync Delays: Sensor data delays may be caused by poor rural bandwidth, which prevents the "Optimize communication speed" aim from being met.
2	External Weather API Outages: The "External Weather API" supplier may either hit the rate restrictions or not responding at all.
3	Wi-Fi to SIM Network Switching Failures: When Wi-Fi fails, sensors may be unable to smoothly switch to the SIM network ("Switch between wifi and sim").
4	Excessive Power Consumption in Sensors: Unexpected power usage or low-power mode failure may quickly drain edge node batteries and disrupt irrigation operations.
5	Database Schema Incompatibility: Inconsistencies between incoming telemetry data types and MongoDB collections during the "Sensor Node Database Mapping" process.
6	Local Weather Forecasting Deviations: Discrepancies between the actual field circumstances and the microclimate data produced by the "Local Weather Forecasting" module.
7	Digital Twin Testing Difficulties: Difficulties in replicating "Virtual Twin Data Matching" tests in a lab setting without a complete field setup.
8	3D Model Render Performance Issues: The "3D Field Model Rendering" may slow down the interface by using too much RAM and CPU on client devices.
9	GPS Positioning Deviations: During "GPS-Based Node Placement," hardware limitations may cause nodes to appear 5–10 meters off the map.
10	Autonomous Irrigation Algorithm Errors: The system opens valves at the incorrect time due to logical errors in the "Irrigation decision logic" (over/under-watering).
11	Industrial Hardware Shielding Degradation: Despite "Industrial Hardware Shielding," hardware can still sustain physical damage from intense heat, ice, or acidic fertilizers.
12	Weekly Summary Script Formatting Glitches: The "Weekly summary script" outputs sometimes may have visual layout issues or email dispatch delays that result in reports that are not correct.

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IMPACT RANK	RISK DESCRIPTION
1	Autonomous Irrigation Algorithm Errors: The system opens valves at the incorrect time due to logical errors in the "Irrigation decision logic" (over/under-watering).
2	Industrial Hardware Shielding Degradation: Despite "Industrial Hardware Shielding," hardware can still sustain physical damage from intense heat, ice, or acidic fertilizers.
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LIKELIHOOD RANK	IMPACT RANK	COMBINED RANK	RISK DESCRIPTION
1	4	5	Communication Speed and Sync Delays: Sensor data delays may be caused by poor rural bandwidth, which prevents the "Optimize communication speed" aim from being met.
3	3	6	Wi-Fi to SIM Network Switching Failures: When Wi-Fi fails, sensors may be unable to smoothly switch to the SIM network ("Switch between wifi and sim").
4	5	9	Excessive Power Consumption in Sensors: The "Low power consumption mode" optimization may fail, resulting in the fast depletion of edge node batteries.
2	7	9	External Weather API Outages: The "External Weather API" supplier may either hit the rate restrictions or not responding at all.
10	1	11	Autonomous Irrigation Algorithm Errors: The system opens valves at the incorrect time due to logical errors in the "Irrigation decision logic" (over/under-watering).
5	6	11	Database Schema Incompatibility: Inconsistencies between incoming telemetry data types and MongoDB collections during the "Sensor Node Database Mapping" process.
11	2	13	Industrial Hardware Shielding Degradation: Despite "Industrial Hardware Shielding," hardware can still sustain physical damage from intense heat, ice, or acidic fertilizers.
6	9	15	Local Weather Forecasting Deviations: Discrepancies between the actual field circumstances and the microclimate data produced by the "Local Weather Forecasting" module.
7	8	15	Digital Twin Testing Difficulties: Difficulties in replicating "Virtual Twin Data Matching" tests in a lab setting without a complete field setup.
9	10	19	GPS Positioning Deviations: During "GPS-Based Node Placement," hardware limitations may cause nodes to appear 5–10 meters off the map.
8	11	19	3D Model Render Performance Issues: The "3D Field Model Rendering" may slow down the interface by using too much RAM and CPU on client devices.
12	12	24	Weekly Summary Script Formatting Glitches: The "Weekly summary script" outputs sometimes may have visual layout issues or email dispatch delays that result in reports that are not correct.